

Homework 1

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Problem 1

Problem Statement

Let a and b be integers, Prove that if $2 + a$ and $35 - b$ are divisible by 11, then $a + b$ is divisible by 11.

Solution

Proof. Since $2 + a$ and $35 - b$ are divisible by 11, it can be represented with the following equations:

$$\begin{aligned}2 + a &= 11 \cdot p, & p &\in \mathbb{Z} \\35 - b &= 11 \cdot q, & q &\in \mathbb{Z}\end{aligned}$$

where p and q are integers.

Each equations can be rearranged as follows.

$$\begin{aligned}a &= 11 \cdot p - 2 \\b &= 35 - 11 \cdot q\end{aligned}$$

The sum $a + b$ is then as follows.

$$\begin{aligned}a + b &= 11p - 2 + 35 - 11q \\&= 11 \cdot (p - q) + 33 \\&= 11 \cdot (p - q) + 11 \cdot 3 \\&= 11 \cdot (p - q + 3) \\&= 11 \cdot k & k = p - q + 3 \in \mathbb{Z}\end{aligned}$$

k is guaranteed to be an integer since it is a sum of integers. Since $a + b$ is a product of 11 and some integer k , $a + b$ is divisible by 11.

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Problem 2

Textbook p15 (15.c, 15.e)

Problem Statement

Use Euclidean algorithm to find:

(a) $(1003, 456)$

(b) $(5858, 1436)$

Solution

(a)

$$\begin{aligned}(1003, 456) &= (456, 91) \\ &= (91, 1) \\ &= (1, 0) \\ &= 1\end{aligned}$$

$$\begin{aligned}1003 &= 456 \times 2 + 91 \\ 456 &= 91 \times 5 + 1 \\ 91 &= 1 \times 91 + 0\end{aligned}$$

Answer: $(1003, 456) = 1$

(b)

$$\begin{aligned}(5858, 1436) &= (1436, 114) \\ &= (114, 68) \\ &= (68, 46) \\ &= (46, 22) \\ &= (22, 2) \\ &= (2, 0) \\ &= 2\end{aligned}$$

$$\begin{aligned}5858 &= 1436 \times 4 + 114 \\ 1436 &= 114 \times 12 + 68 \\ 114 &= 68 \times 1 + 46 \\ 68 &= 46 \times 1 + 22 \\ 46 &= 22 \times 2 + 2 \\ 22 &= 2 \times 11 + 0\end{aligned}$$

Answer: $(5858, 1436) = 2$

Problem 3

Textbook p15 (15.j)

Problem Statement

Express $(1003, 456)$ as a linear combination of 1003 and 456.

Solution

From using the Euclidean algorithm in Problem 2 (a), following equations are found.

$$1003 = 456 \times 2 + 91 \quad (1)$$

$$456 = 91 \times 5 + 1 \quad (2)$$

These equations can be rearranged as follows.

$$91 = 1003 - 456 \times 2 \quad (3)$$

$$1 = 456 - 91 \times 5 \quad (4)$$

By substituting Equation (3) to Equation (4), 1 can be represented as a linear combination of 1003 and 456.

$$\begin{aligned} 1 &= 456 - (1003 - 456 \times 2) \times 5 \\ &= 456 \times 11 - 1003 \times 5 \end{aligned} \quad (5)$$

The GCD of 1003 and 456 is 1, which can be represented as a linear combination of 1003 and 456 as follows.

$$1 = 456 \times 11 - 1003 \times 5$$

Problem 4

Problem 5

Problem 6

Problem 7